

《Automotive Configuration》

Chapter 8

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第八章 发动机润滑系

Chapter 8 Lubrication System for Automotive Engines

8.1 Lubrication system and its functions

润滑系统的功用与组成

8.2 Lubricants

润滑剂

8.3 Oil pumps

机油泵

8.4 Oil filters

机油滤清器

8.5 Heat transmitters (Oil cooler)

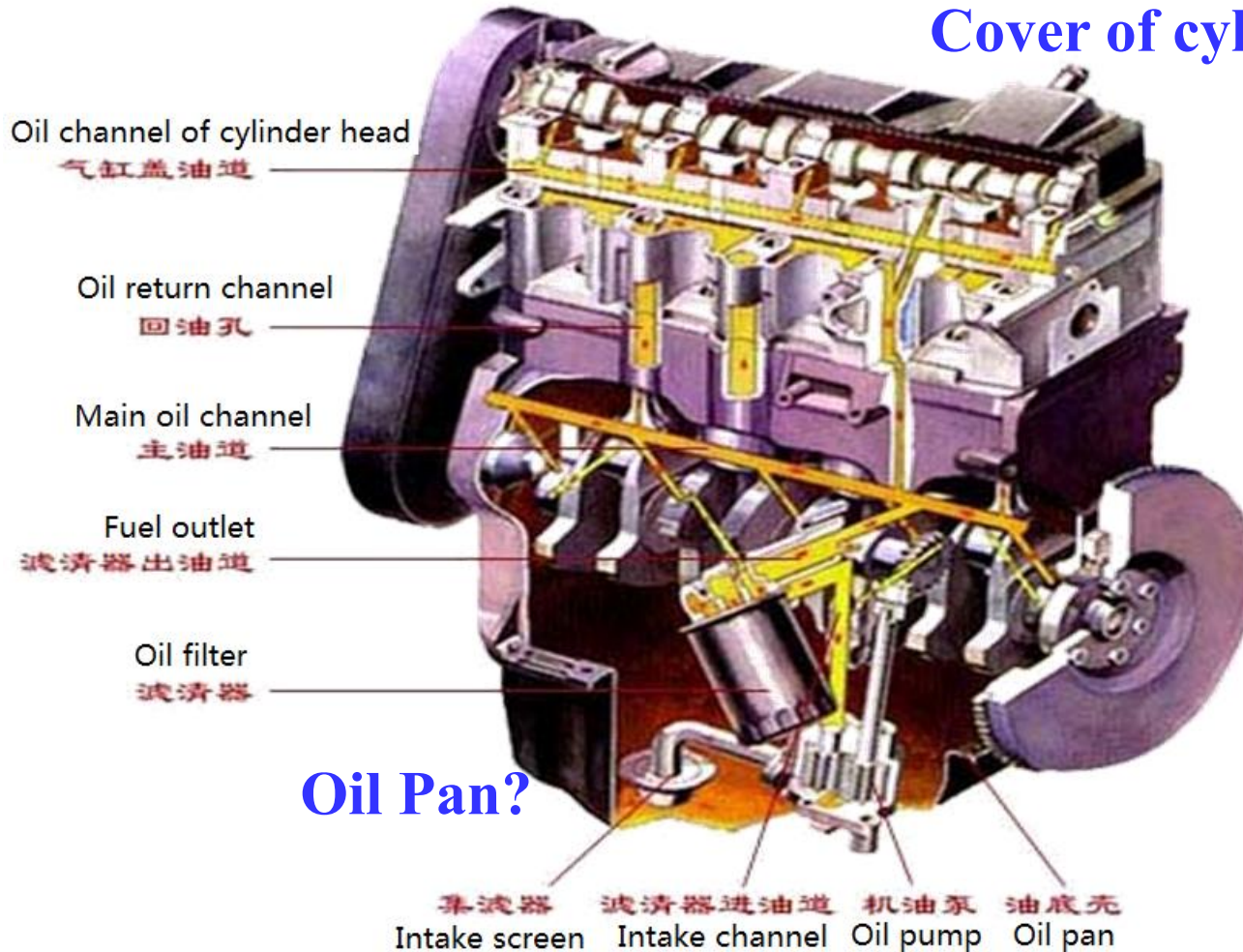
机油冷却器

Oil?
Fuel?

8.1 Lubrication system and its functions

(润滑系统组成及功用)

Cover of cylinder head?



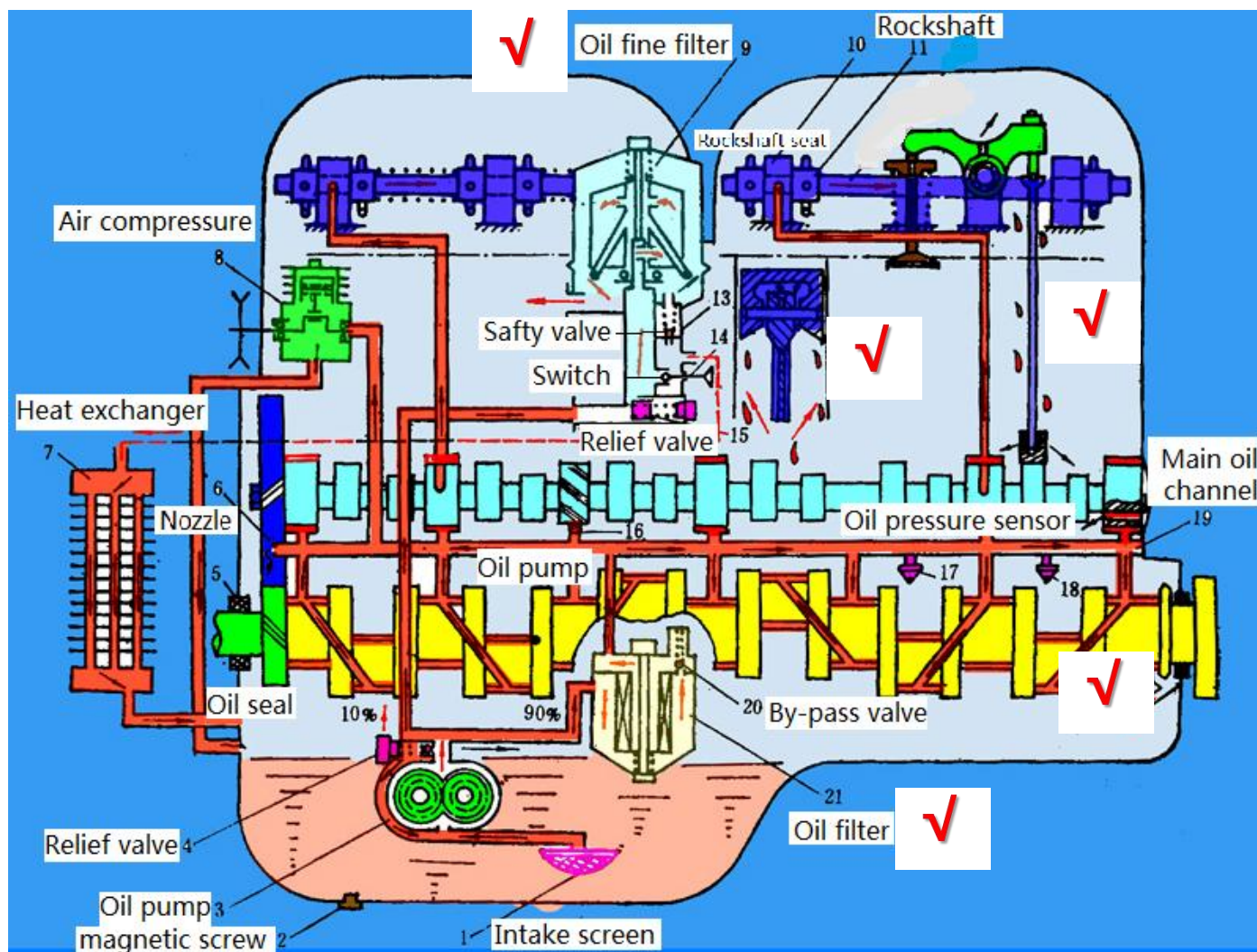
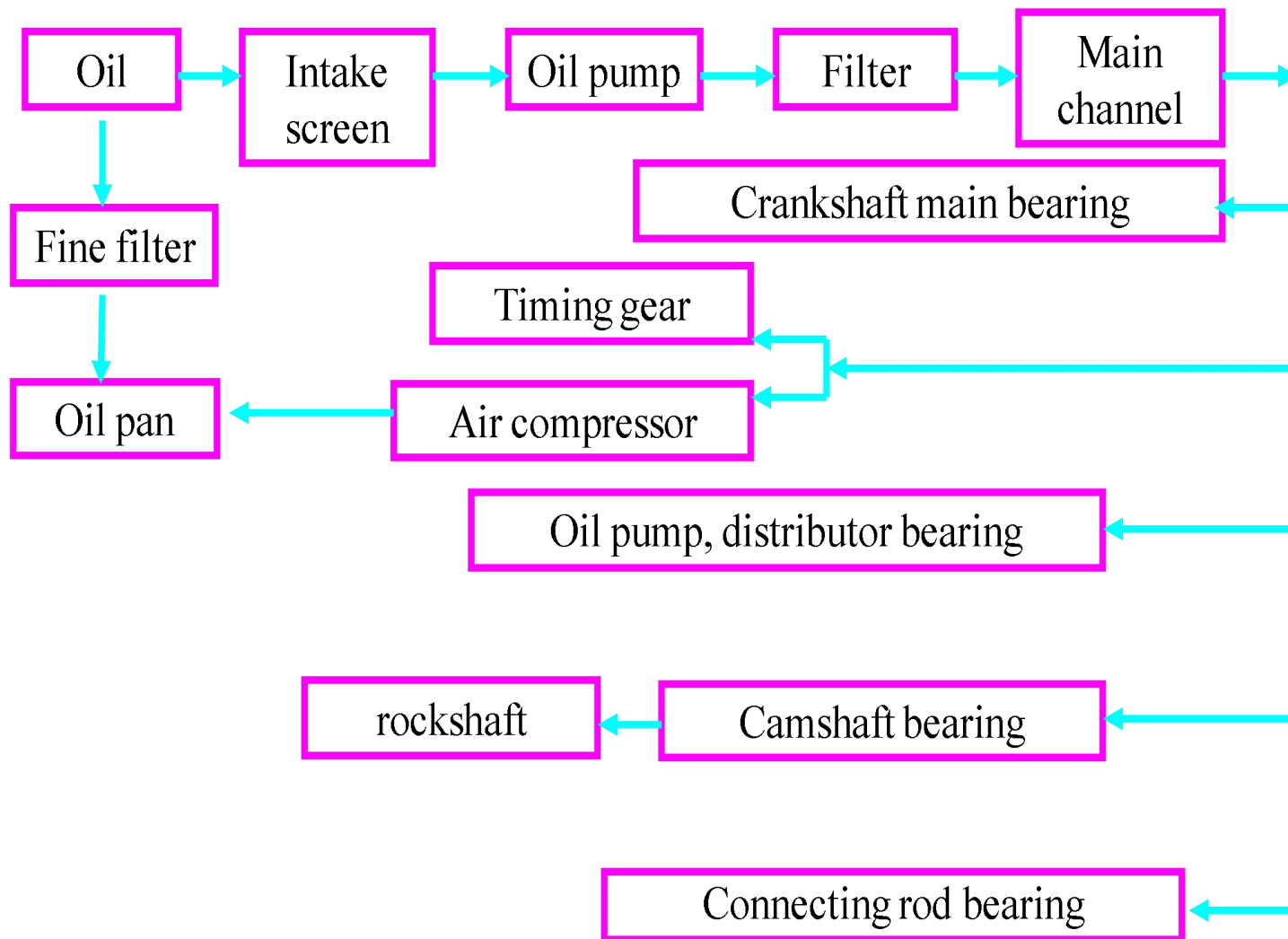


图 6-4 汽油发动机润滑油路示意图

1-集滤器;2-磁性放油螺塞;3-机油泵;4-限压阀;5-曲轴前油封;6-喷嘴;7-机油散热器;8-空气压缩机;9-细滤器;10-摇臂轴支座;11-摇臂;12-摇臂轴;13-机油散热器安全阀;14-机油散热器开关;15-机油限压阀;16-机油泵和分电器驱动轴;17-油压过低传感器;18-油压传感器;19-主油道;20-旁通阀;21-粗滤器



Main lubricating methods in engines

- **Pressurized Lubrication (Forced-feed lubrication)** 压力润滑
 - ✓ Oil is pumped to friction surfaces at certain pressure
 - ✓ Used in **crank bearings, cam bearings, etc**
- **Splash lubrication** 飞溅润滑
 - ✓ Oil is splashed to friction surfaces by movement of engine components
 - ✓ Used for friction surfaces that exposed to air, such as **cylinder liner, piston pin, cam face, and tappet**
- **Grease Lubrication (Periodic lubrication)** 脂润滑
 - ✓ Greases is used instead of oil to do periodic lubrication for engine components with light loads
 - ✓ Mainly used in **bearings for water pump and generator**

Functions

- ✓ Reduction of friction and wear (润滑作用)
- ✓ Anti-corrosion (防锈作用)
- ✓ Cooling (冷却作用)
- ✓ Cleaning (清洗作用)
- ✓ Sealing (密封作用)
- ✓ Power transmission (传动作用)
- ✓ Reduction of noise (降噪作用)
- ✓ Damping impact and vibration (减震缓冲作用)

8.2 Lubricants (润滑剂)

- Types of lubricants

- ✓ Lubrication oils

- Engine oils (机油)
- Universal oils for tractors (通用拖拉机油)
- Oils for two-stroke motorcycles (二冲程摩托车用机油)
- Transmission oils (传动润滑油)



- ✓ Greases (润滑脂)



- Task of lubricants

- ✓ Wear protection (润滑)
- ✓ Cooling (冷却)
- ✓ Protection from deposits (清洗积碳)
- ✓ Sealing (密封)
- ✓ Corrosion protection (防锈)

8.2.1 Lubricant Characteristics and Additives

(润滑剂特性及润滑剂添加剂)

- **Working condition**

- ✓ Exposed to high temperature air and metal walls, easily get oxidized
- ✓ Severely polluted by fuel vapor, exhaust gas, wear particles and deposits
- ✓ Wide range of working temperature (ambient temperature to over 300°C)

- **Lubrication Requirements**

- ✓ Viscosity Temperature Behavior: thin when cold, viscous when hot (适当的粘度)
- ✓ Aging stability: resist of oxidation (氧化安定性)
- ✓ Anti-corrosion property: oxidation of oil generates acid (防腐蚀性)
- ✓ Low foamability (低起泡性)
- ✓ Detergency and dispersibility: ability to remove wear particles and deposits (清净分散性)
- ✓ Extreme pressure property: pressure in the thin lubrication gap can be 10000bar (高压压性)

Lubrication Oil {

Basic liquids



Mineral Basic Oils (矿物质油) : conventionally used

Synthetic Basic Liquid (合成油) : high-performance multigrade oil

Additives for lubricants (see table below)

Additive type	Function
V.I. improvers (粘度改善剂)	Improve viscosity temperature behavior
Detergents (清洁剂)	Keep engine clean, neutralize acids
Dispersants (分散剂)	Disperse soot, aging products, foreign materials, prevent deposits
Oxidation inhibitors (抗氧化剂)	Prevent oil oxidation and thickening
Corrosion inhibitors (抗腐蚀剂)	Prevent corrosion
Nonferrous metal deactivators (有色金属抗活化剂)	Prevent oxidation and oil thickening
Friction modifiers (摩擦改进剂)	Reduce friction loss
Wear reducer (EP additives) (抗磨损剂)	Reduce or prevent wear
Pour point depressant (降凝剂)	Improve flow properties at low temperatures
Foam inhibitor (泡沫抑制剂)	Reduce or prevent the formation of foam

8.2.2 Classification of Lubrication Oil (润滑油分类)

- SAE (Society of Automotive Engineers) introduced a binding classification for engine oil viscosity (1911)
 - ✓ A total of 12 classes:
 - { Six for the winter: SAE0W-SAE25W
 - Six for the summer: SAE20-SAE60
 - ✓ Higher numbers indicate higher viscosity and is suitable for higher engine speed
 - ✓ Engine oils used today normally cover the viscosity requirements of several SAE grades, and is indicated as “multigrade oil”(多粘度等级), such as SAE5W/20, SAE10W/30
 - ✓ SAE5W/20 has viscosity of SAE5W in winter and viscosity of SAE20 in summer, therefore could be used throughout whole year

- **API (American Petroleum Institute) classifies lubrication oil according to its performance and appropriate applications**
 - {** S series: for gasoline engines, classified from SA to SN
 - {** C series: for diesel engines, classified from CA to CE (Performance increases alphabetically)
- **Domestic classification adopts ISO method according to GB/T7631.3-1995**
 - {** Gasoline engines: classified from SC to SH
 - {** Diesel engines: classified as CC, CD, CD-2, CE, CF-4
 - {** Two stroke gasoline engines: ERA to ERD
- **Greases (润滑脂)**
 - ✓ By mixing lubrication liquids with densifier, one can get a solid or semisolid product
 - ✓ Stable and immobile(不可流动), suitable for open type or sealless (无密封) friction parts

8.3 Oil Pump (机油泵)

- **Function:** Oil delivery, pressure boosting

(供油、增压)

- **Classification:**

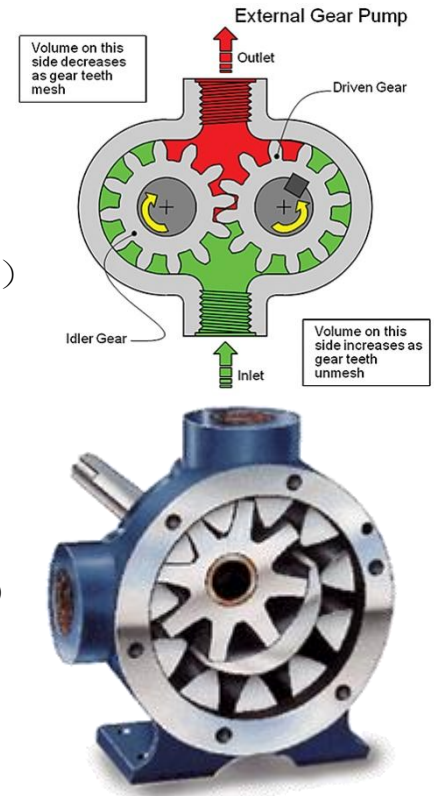
Oil Pump

Gear Pump (齿轮泵)

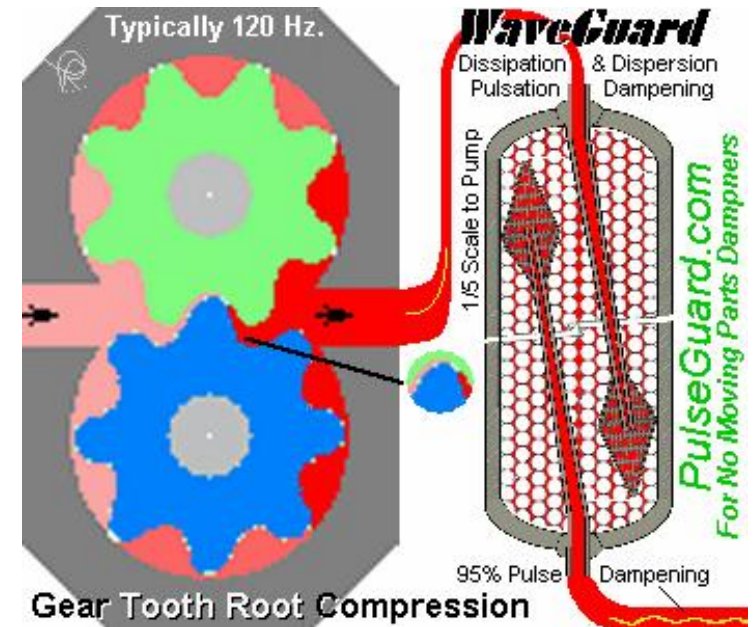
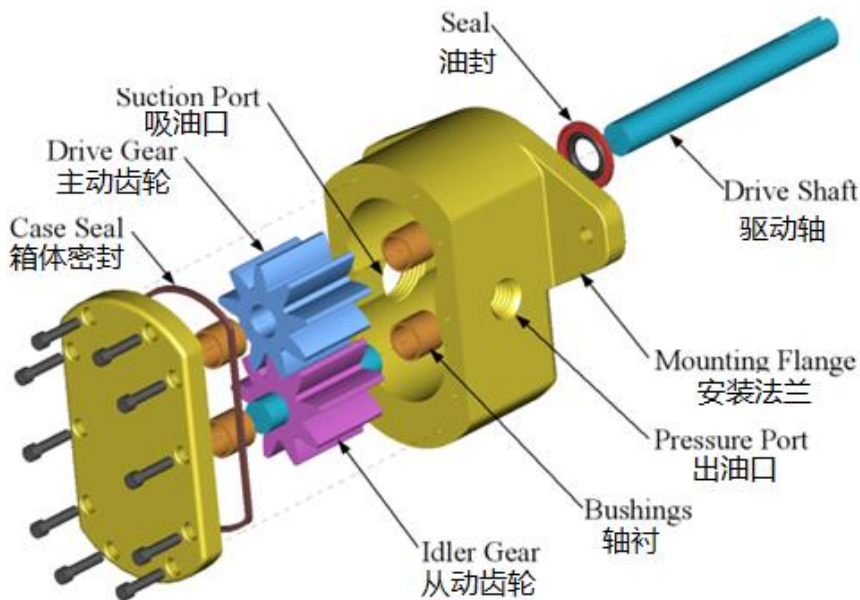
External gear pump (外接齿轮式)

Internal gear pump (内接齿轮式)

Rotor Pump (转子式)

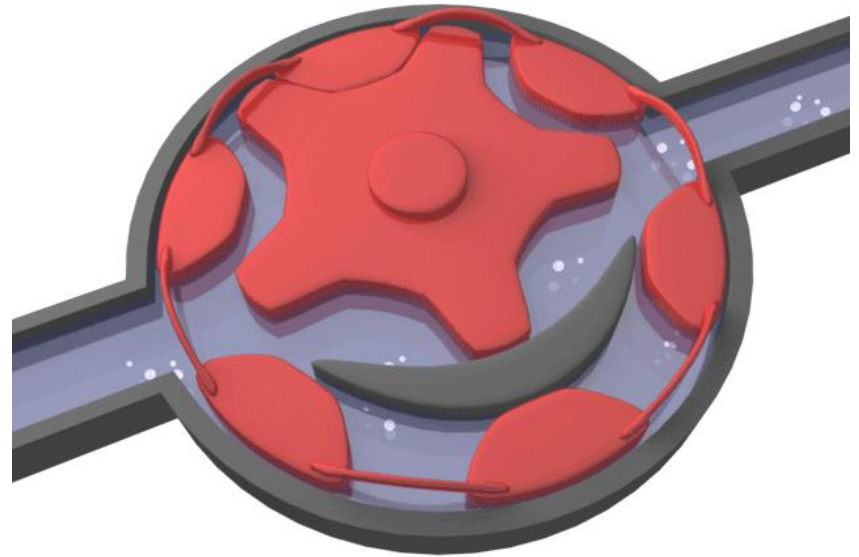
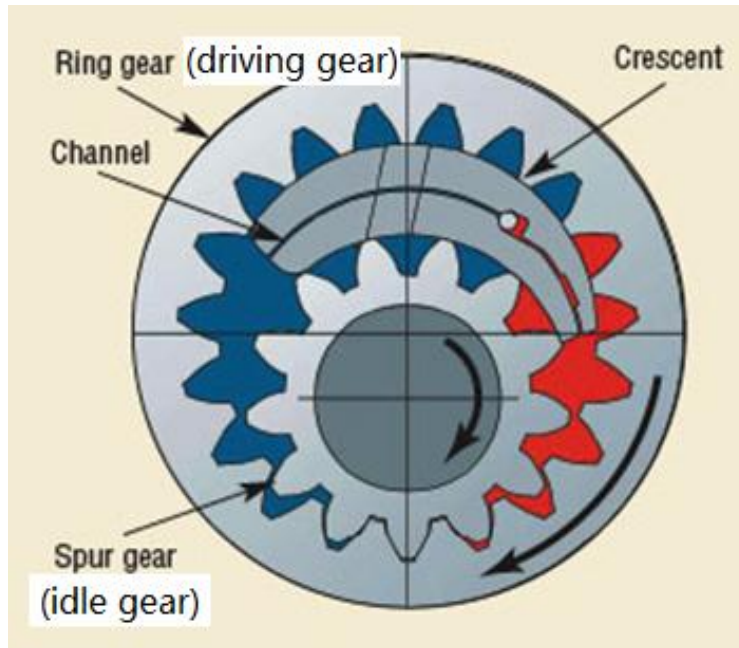


8.3.1 External Gear Pump (外接式齿轮泵)



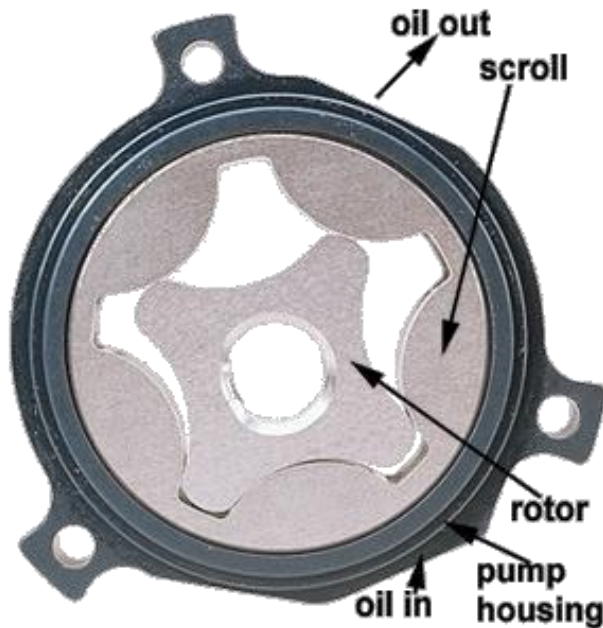
- **Advantages:** high efficiency, simple structure, reliable, low power loss
- **Disadvantages:** requires intermediate transmission (中间传动装置), larger size, high manufacture cost, uneven oil supply (供油不均匀)

8.3.2 Internal Gear Pump (内接式齿轮泵)



- **Advantages:** No intermediate transmission, less components, lower cost, smaller size
- **Disadvantages:** low volume efficiency (because of the crescent (月牙形)), higher power loss

8.3.3 Rotor Pump (转子泵)



- **Advantages:** Compact structure, smaller size, less weight, even oil supply, lower cost , widely used in small engines with light load



可变排量机油泵



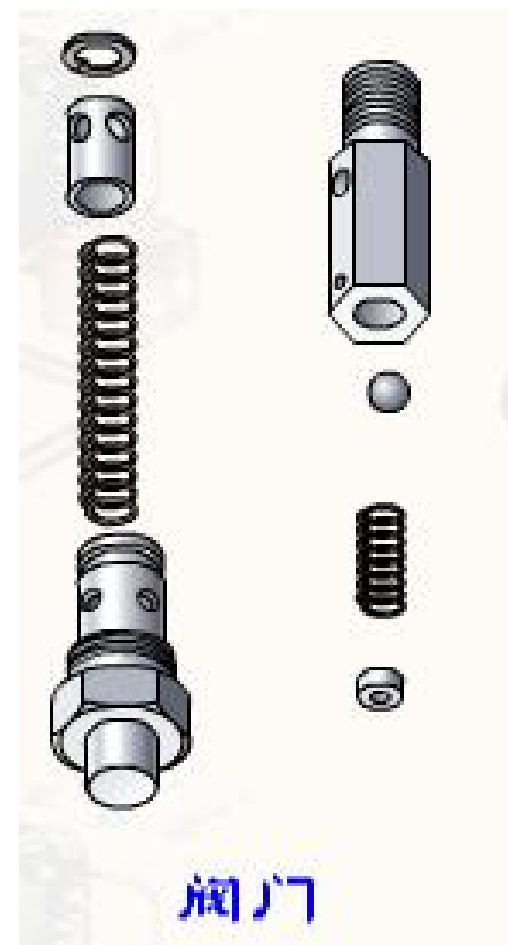
电动机油泵

- **Relief Valve (限压阀)**

- ✓ High oil pressure would lead to engine power loss
- ✓ Relief valve is placed on oil pump cap (油泵端盖)
- ✓ Relief valve opens when oil pressure greater than 0.6MPa, and let part of oil goes back to oil inlet, and close when oil pressure lower than 0.1MPa

- **Bypass Valve (旁通阀)**

- ✓ Placed in parallel with (并联) full-flow filter (细滤器) between oil pump and main oil channel
- ✓ When filter is blocked up, pressure difference between inlet and outlet of filter would rise
- ✓ When pressure difference reaches 0.15MPa, bypass valve opens to guarantee enough oil delivered to engine



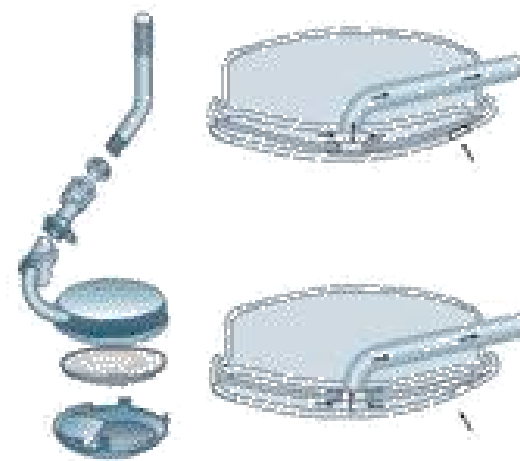
8.4 Oil Filter (机油滤清器)

Oil filters:

- ✓ Intake screen (集滤器)
- ✓ Full-flow oil filters (粗滤器): in series with (串联) main oil channel
- ✓ Bypass oil filters (fine filter)(细滤器): in parallel with (并联) main oil channel, bypass 10~30% oil

- **Intake screen**

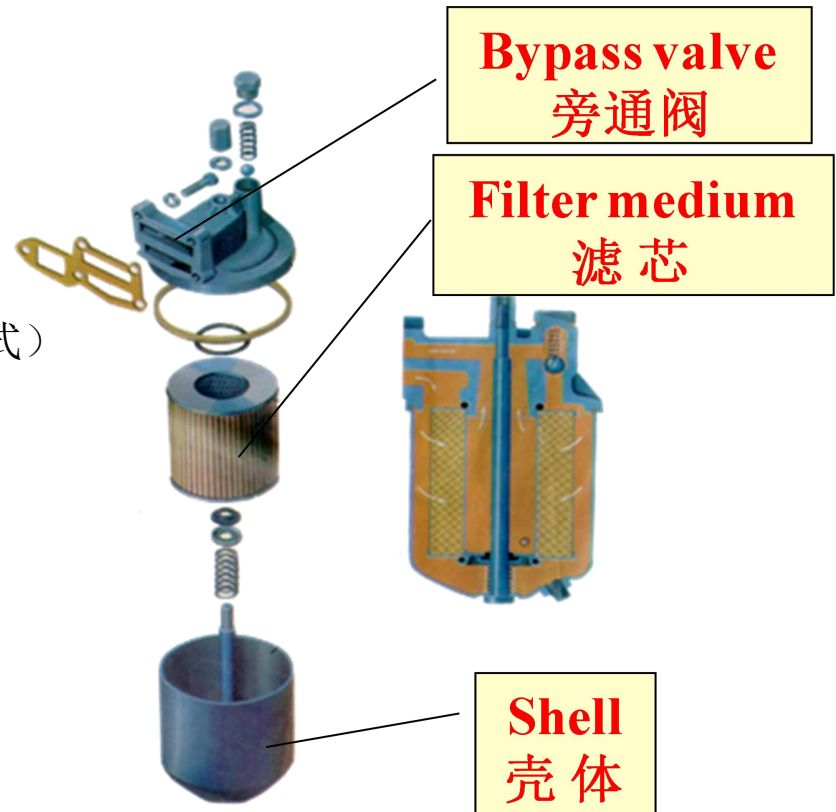
- ✓ Prevent large metallic particles into the oil pump
- ✓ **floating type** (浮式): floating on oil surface, draws cleaner oil at the top, lower oil pressure because of foam (泡沫) suction
- ✓ **Stationary type** (固定式): placed under oil surface, prevent foam drawing, more reliable, simpler structure

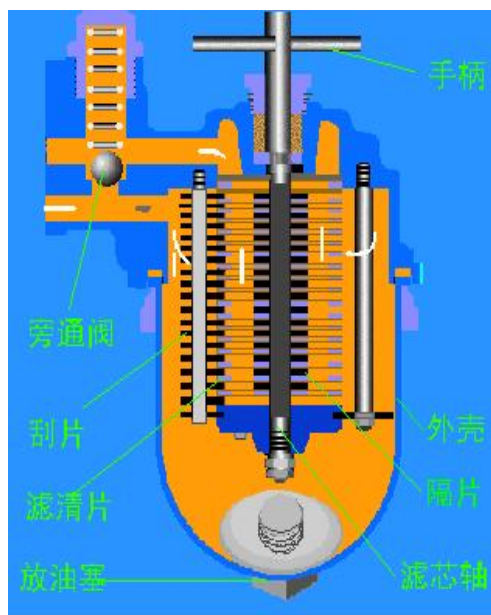


集滤器

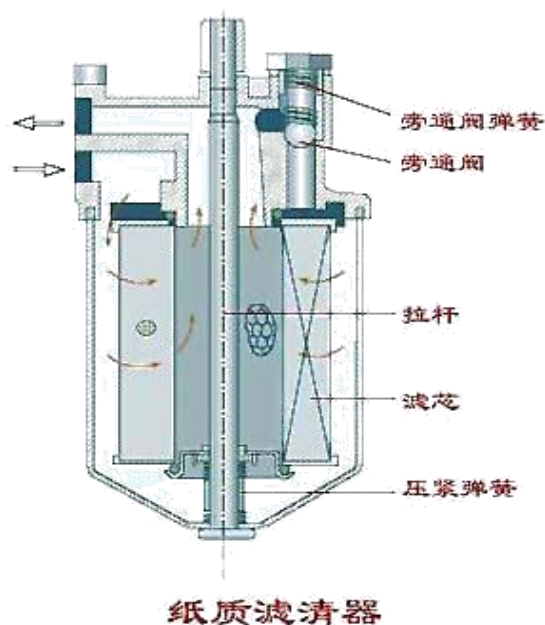
8.4.1 Full-Flow Oil Filters (粗滤器)

- In series (串联) with oil pump and main oil channel
- Types:
 - Metal elements (金属片式)
 - Paper elements (纸筒式)
 - Saw powder elements (锯末式)
- Bypass valve works in case filter element is block up and no oil is delivered to engine

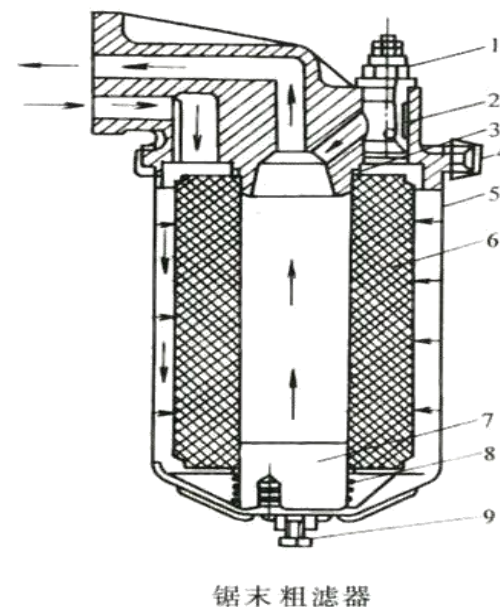




- Hardly used because of its weight, complex structure, and production cost



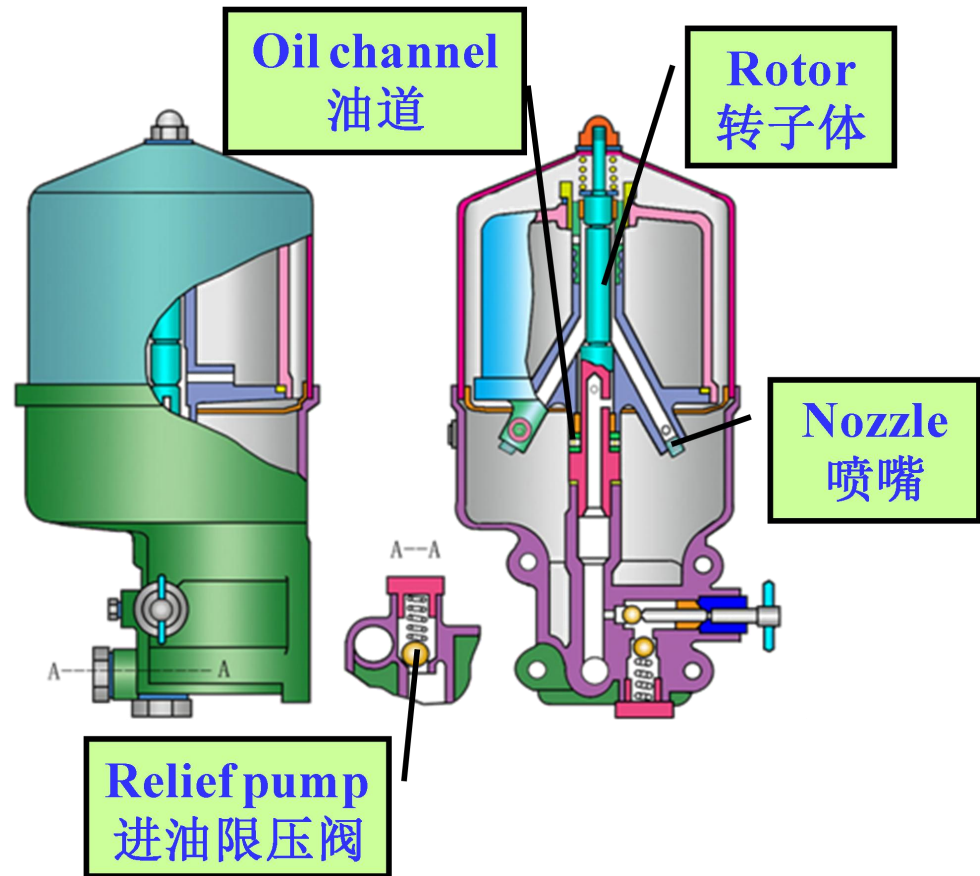
- Most widely used
- Light, compact, simpler structure with low cost
- Low filtration resistance (过滤阻力)
- Easy maintenance



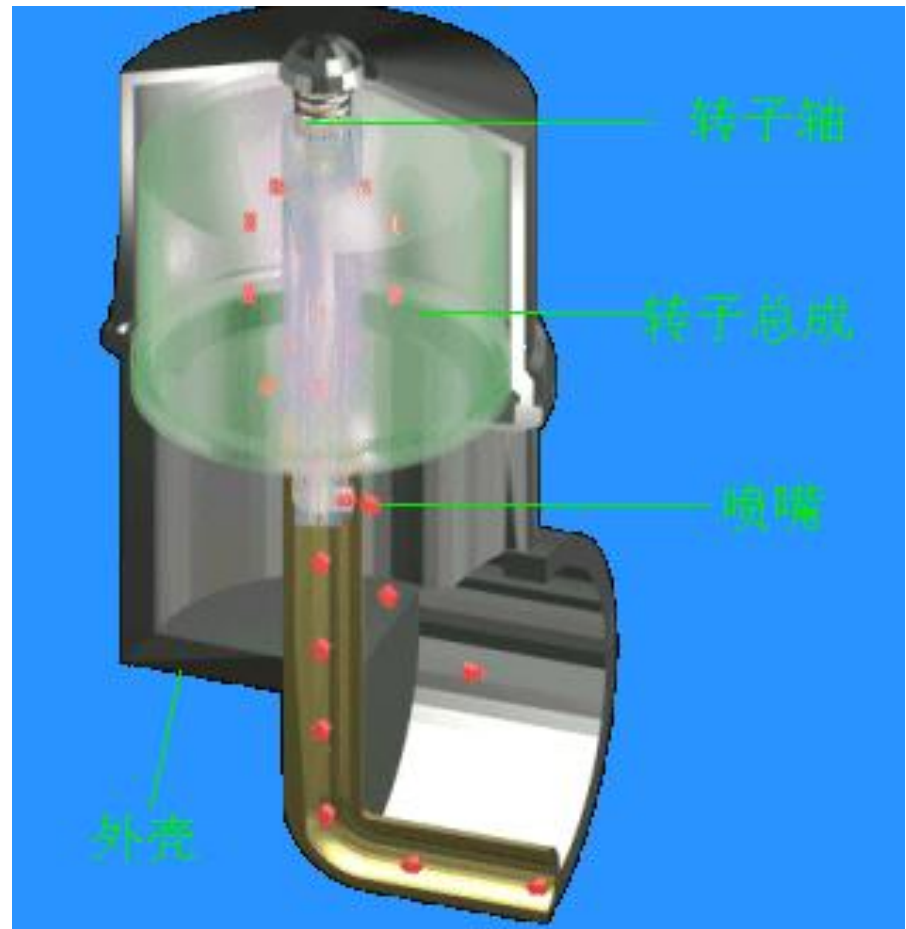
- Composite filter elements (复合材料滤芯)
- Low filtration resistance
- Long service life

8.4.2 Bypass Oil Filters (Fine Oil Filter) (细滤器)

- Removing element for ultrafine particles (超微颗粒)
- Particularly for high-mileage diesel engines with high soot level
- In parallel with main oil pass (high filtration resistance)
- Classification
 - ✓ Net-type filter: Dilemma between filtration capacity (滤清能力) and through capacity (通过能力)
 - ✓ Centrifugal filter: most widely used

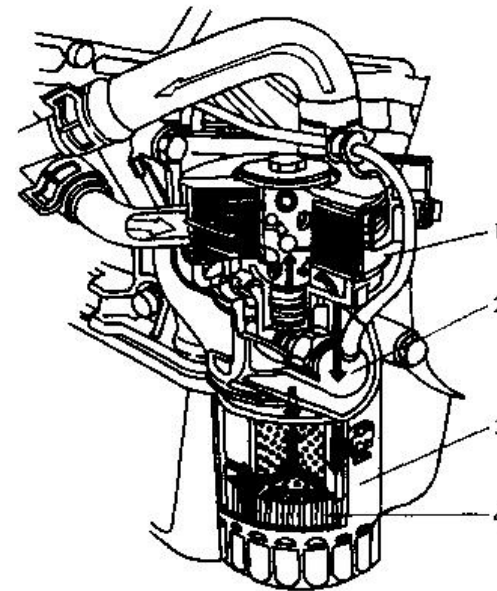
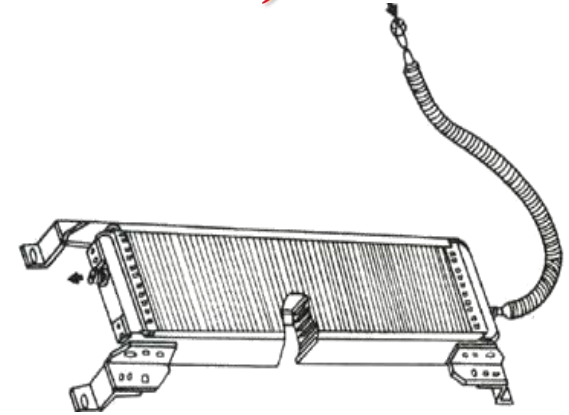


- Consist of metal or plastic
- Rotation speed up to 10000 r/min
- Extremely compact filter cake (滤饼), easy maintenance



8.5 Heat Transmitters (Oil cooler) (机油冷却器)

- Control oil temperature within appropriate range at high engine loads
- Forced cooling device
- Classification
 - ✓ Wind cooled: Placed before engine radiator, making use of the cooling fan
 - ✓ Water cooled: Placed in engine cooling channel
- Gain heat from cooling water during cold start
- Dissipate heat to cooling water under high load
- Compact structure, widely used in passenger cars
- Lubrication oil goes into heat transmitter after it leaves oil filter



- 1、Heat transmitter
机油冷却器
- 2、Pressure switch
机油压力开关
- 3、Oil filter
机油滤清器
- 4、Filter element
机油滤清器滤芯